Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/09/2020
PI Name:	Wold, Barbara Ph.D.		
Project Title:	Discovery and Definition of the Microgravity Transcriptome, with Application to Muscle Atrophy and Regeneration Processes		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	03/01/2002	**End Date:**	05/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE to 5/31/2006, per A. Chu 7/2006		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shapiro, Bruce (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	NAG2-1540		
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	The project has integrated goals in development and application of bioinformatic tools for storing, analyzing and mining large scale gene expression data and comparative genomic data. Once primary data analysis is complete, it is integrated with andditional public genomic, proteomic, and functional annotation data. The informatics tools are thus applied to wet-bench microarray measurements using material from microgravity models. Project Aims: To develop bioinformatics tools that will be publicly available for use by the community. To use these and additional tools to discover the cellular pathways and circuits that control muscle atrophy, with		

Task Description:	emphasis on microgravity paradigms. Alterations in gravity impact cells in many ways, including cell proliferation, chromosomal aberrations, gene expression, and processes of reproductive cell formation. At the systemic level, microgravity affects virtually every system in the body. Known detrimental changes include bone mineral density losses that vary across specific bone sites with greater loss in the weight-bearing bones. Marked atrophy of postural muscles as well as reduced muscle strength and tasks prevent astronauts from performing sustained work in space and cause severe side-effects upon return to Earth. These effects increase with time in flight and it is not known how long they continue to develop. This project will help to understand the molecular and cellular causes of muscle atrophy and identify potential targets for countermeasure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth benefits from this work are in these areas: 1) The bioinformatics and analysis tools are applicable to all microarray data including vast array of medical analysis and diagnostics using genomic methods. 2) Microgravity models and medical problems of muscle and bone degeneration associated with, for example, extended bedrest, make the identification of genes involved in these pathways of significance to a large population of patients confined to bedrest.
Task Progress:	[ED. note January 2020: compiled from final report received from NASA Ames Research Center (ARC) personnel in August 2019; received by ARC in late 2007] Rat hindlimb unloading samples: Skeletal muscle samples of soleus, gastrocnemius, and tibialis anterior from rats on a one week hindlimb unloading protocol were processed for total RNA extraction, along with samples of left cardiac ventricle. Total RNA yield per milligram of tissue mass was significantly reduced for soleus and gastrocnemius compared to ambulatory control rats, confirming previously published results. Total RNA samples were reverse transcribed into double stranded cDNA containing a T7 promoter site, from which biotin labeled cRNA was produced. cRNA samples were analyzed for consistent yield and length distribution by electropherogram, prior to hybridization to Affymetrix Rat Expression set 230 version 2.0 chips. Each chip assays 30,248 expressed sequences and variants using 11 probe pairs per sequence. The average portion of chip capacity reliably detected in the eight hybridizations performed was 46% ± 4%. In the initial expression analysis, a singular value decomposition method was used to identify eight principal components (PCs) for the eight factor dataset comprised of genes that were reliably detected in both experimental and control soleus muscle, which has previously been shown to be most sensitive to hindlimb unloading. CompClust and PCA Interpretation: CompClust is a Python package for exploratory data analysis. A novel principal components analysis (PCA)-based tool was developed under this effort to extend CompClust use. It has now become the primary tool that other biologist colleagues have been using to analyze human, mouse, and rat microarray expression data in order to identify subsets of co-expressed genes involved in muscle development and atrophy pathways. Cistemic Pipeline: Our CalTech collaborators have developed a Python package called Cistematic for efficiently analyzing cis-regulatory elements within and across multiple genomes. BioHub: The Jet Propulsion Laboratory (JPL)/CalTech collaboration has been developing a system called BioHub to support integrated analysis of diverse sequence annotation and transcript expression data. BioHub is a sequence annotation database and Python API that manages associations between numerous genomic-sequence-based and transcript-based databases in order to provide centralized query services and uniform data access. Sigmoid: The Sigmoid Project is intended to produce a database of cellular signaling pathways and models, including the major forms of data and knowledge required as input to cellular modeling software and to organize outputs. Cellerator: Cellerator is a Mathematica package for describing signal transduction networks. Enhancements funded under the current project include the addition of reversible Michaelis-Menten kinetics, Savegeau Systems, and generalized Monod-Wyman-Changeaux allosteric dynamics, and enhanced mass-action-based enzyme kinetics. Details and sample notebooks are available at http://www.cellerator.info MathSBML: this is a Mathematica package that allows users to read and write SBML models into Mathematica data structures. No funds from this project were used for development; however, MathSBML was used by the enhancements to Cellerator for SBML support. Experimental and Genomic Data Analysis: The broadest goal of our collaboration was to define components of the regulatory network underlying skeletal muscle atrophy, using an extensive collection of bioinformatics tools to interpret genome-wide expression data obtained from experimental animal models of the microgravity environment. Skeletal muscle samples of soleus, gastrocnemius, and tibialis anterior from rats and mice subjected to one-week hindlimb unloading were collected for micro-array gene expression experiments. Two genes in particular, Muscle Atrophy F-box (MAFbx) and Muscle RING Finger-1 (MuRF-1), are ubiquitin ligases that are required for muscle atrophy and are used as molecular markers of atrophy in many studies searching for mediators of the atrophy response in skeletal muscle. By identifying and functionally characterizing genomic cis-regulatory elements that regular the MAFbx and MuRF-1 genes, we sought to identify other genes that may have these regulatory sequences using genome-wide searches for consensus/refined binding sites. Our approach involved characterizing the gene expression profiles for each of the skeletal muscles tested, then using Cistematic to identify and refine candidate cis-regulatory elements within conserved regions of genes with similar expression, followed by a genome-wide search for and characterization of other similarly co-regulated genes.
Bibliography Type:	Description: (Last Updated: 01/09/2020)
Articles in Peer-reviewed Journals	Roden JC, King BW, Trout D, Mortazavi A, Wold BJ, Hart CE. "Mining gene expression data by interpreting principal components." BMC Bioinformatics. 2006 Apr 7;7:194. PubMed PMID: 16600052; PubMed Central PMCID: PMC1501050, Apr-2006

Articles in Peer-reviewed Journals	Li L, Roden J, Shapiro BE, Wold BJ, Bhatia S, Forman SJ, Bhatia R. "Reproducibility, fidelity, and discriminant validity of mRNA amplification for microarray analysis from primary hematopoietic cells." J Mol Diagn. 2005 Feb;7(1):48-56. PubMed PMID: 15681474; PubMed Central PMCID: PMC1867503 , Feb-2005